

Unit 1

Chapter 1: Introduction to Statistics and Use in Business

- **Definition:**

The field of statistics is the science of learning from data. Statistical knowledge helps you use the proper methods to collect the data, employ the correct analyses, and effectively present the results. Statistics is a crucial process behind how we make discoveries in science, make decisions based on data, and make predictions.

- **Importance and Scope of Statistics:**

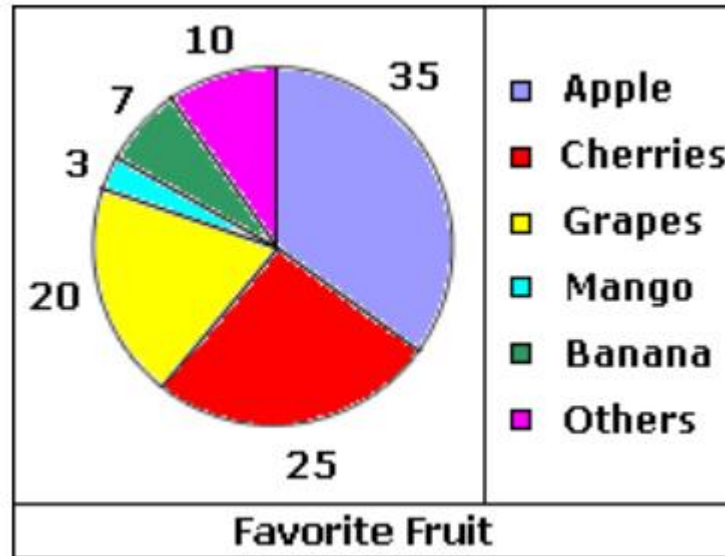
1. **Statistics in Planning :**

- Statistics is one of the crucial parts of planning. Without statistics, the plan can't be possible.
- Statistics helps to do planning in business, economics, government level, and even a person.
- It has been noticed that 63% of people plan their activity a year or less in advance.
- In the modern era, planning is everything.



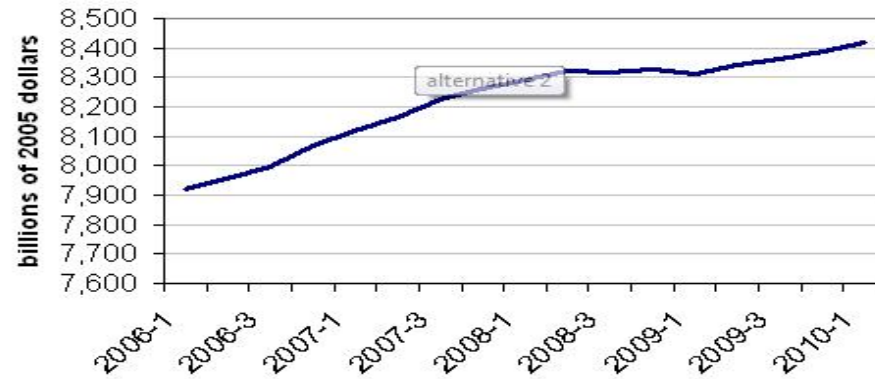
2. Statistics in Mathematics:

- Statistics is an essential part of mathematics. In other words, it is related as well as entirely depends upon mathematics.
- Suppose you want to calculate which is the favourite fruit of your city persons?
- Below is the statistical data that you get using the mean of the mathematical statistics.



- The increasing use of mathematics and statistics is building new ways for the development of statistics, i.e., known as Mathematical statistics.
- Statistics is also crucial for business mathematics.

3. Statistics in Economics:



- Statistics and Economics are interrelated with each other.
- It is impossible to separate them. The development of advanced statistics has open new ways to extensive use of statistics for Economics.
- Almost every branch of Economics uses statistics, i.e., consumption, production, distribution, public finance.
- All these Economic branches use statistics for comparison, presentation, interpretation, and so on.
- Income spending problems on and various sections of the people. National wealth production, demand, and supply adjustment, the effect of economic policies.
- All these indicate the importance of statistics in the field of economics and its various branches.

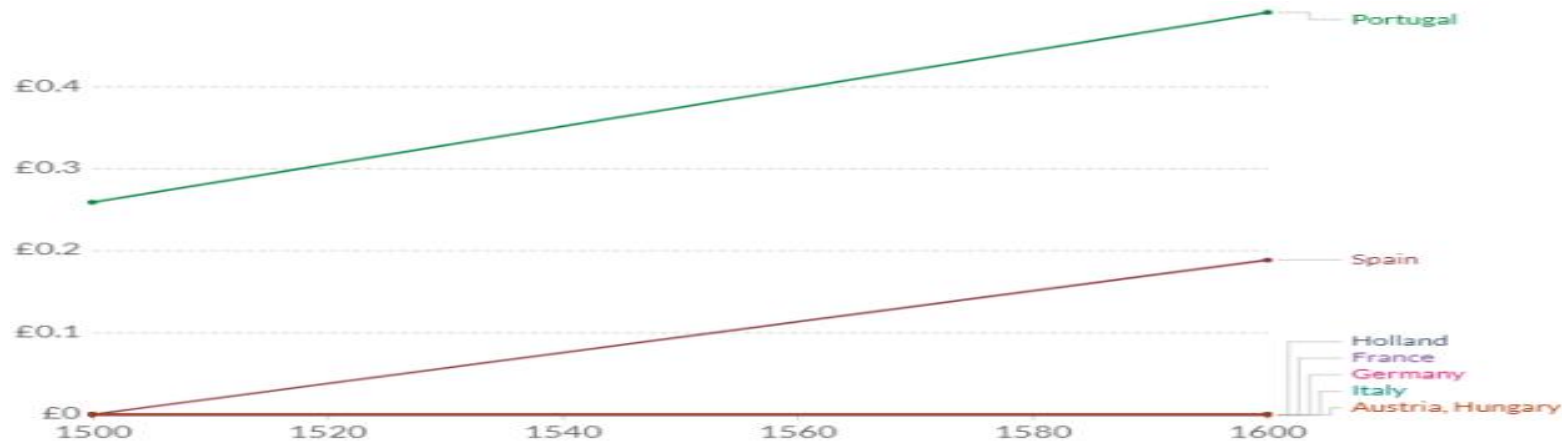
4. Statistics in Social Science



- Social phenomena are affected to a marked and extent by a factor's multiplicity. In social science, the variation in observations from time to time, object to object, and place to place.
- Statistical tools of regression and correlation analysis are used in social science to isolate these factors' effect in every tested observation.
- Statistics are also used to conduct social surveys.
- The social survey uses the Sampling techniques and estimation theory.
- Statistics is also playing a critical role in sociology. It helps in studying mortality, fertility, population growth, and so on.

5. Statistics in Trade

- Trade without statistics is pretty tough and can be overwhelming for traders.
- It helps the trades to make wise decisions in uncertain situations.
- We know that business is full of risks and uncertainties; anything can happen at any stage.
- Below graph, define the international trade per capital of the selected countries

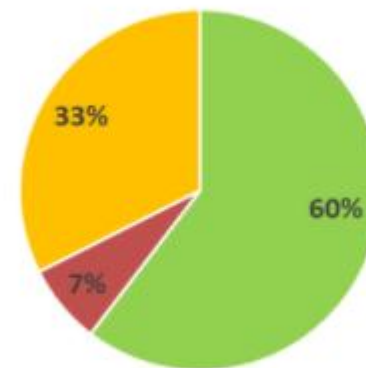


- We take all the recent statistics into account to predict future situations for the company.
- Remember that statistics are just a number, we have to anticipate them carefully; otherwise, it can be a reason for the failure of the business.
- We can anticipate all these aspects with the help of statistics.
- It also helps in determining the prices of various products.
- At last, statistics help in running the business smoothly.

6. Statistics in Research Work

- Research work is all about statistics. The research worker's job is to present the data before the community.
- The research worker uses statistical methods to solve particular problems under differing conditions.
- Statistics are the basics of every research activity. In other words, research can't be possible without statistics.
- The below-mentioned chart is all about the animals used in a research by Pain/Anesthesia category.

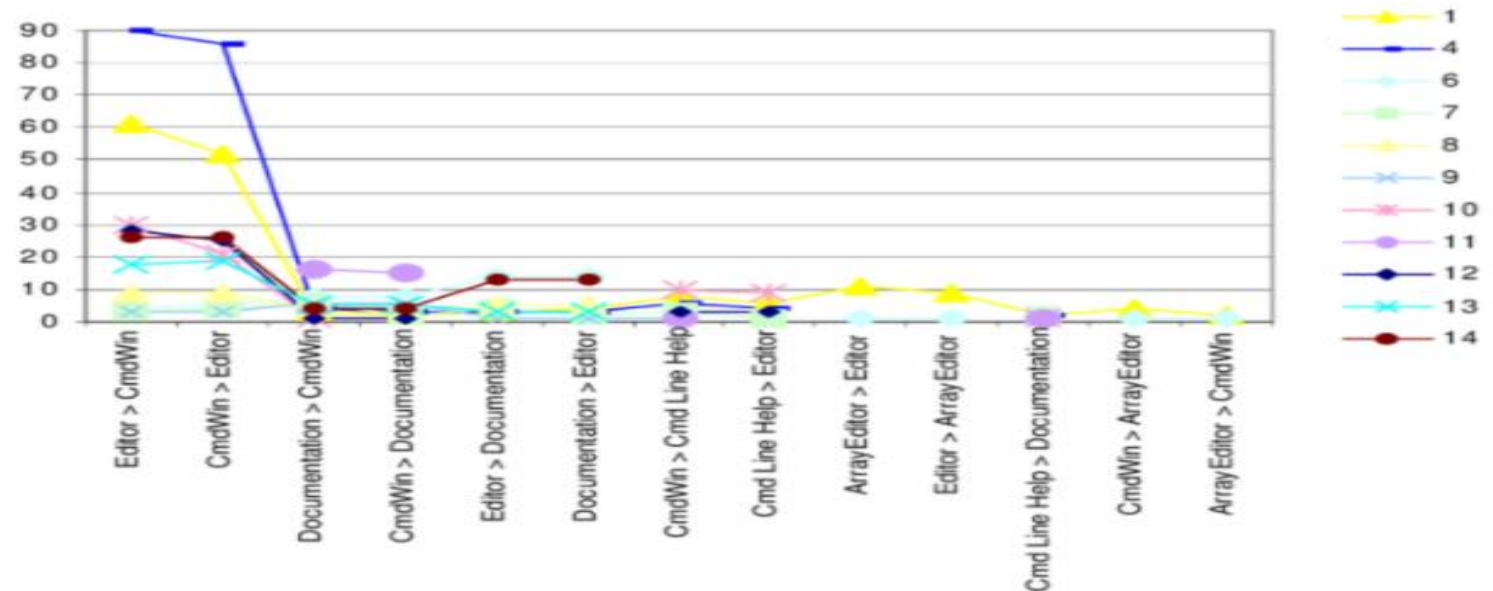
Proportion of Animals used in Research in the US in 2018 by Pain/Anesthesia category



■ No Pain, No Anesthesia ■ Some Pain, No Anesthesia ■ Some Pain, Anesthesia

7. Statistics in Programming

- Nowadays, most of the advanced programming is based on statistics.
- Most of the Python developers are learning statistics to enhance their career in Python programming.
- And did you know that? Statistics is the foundation of artificial intelligence and machine learning.
- You can't get good command over these technologies without the help of statistics.
- It also helps you to improve your programming logic.
- Statistics is widely used in some of the most popular programming languages i.e., Java, Python, Swift, C, and C++.



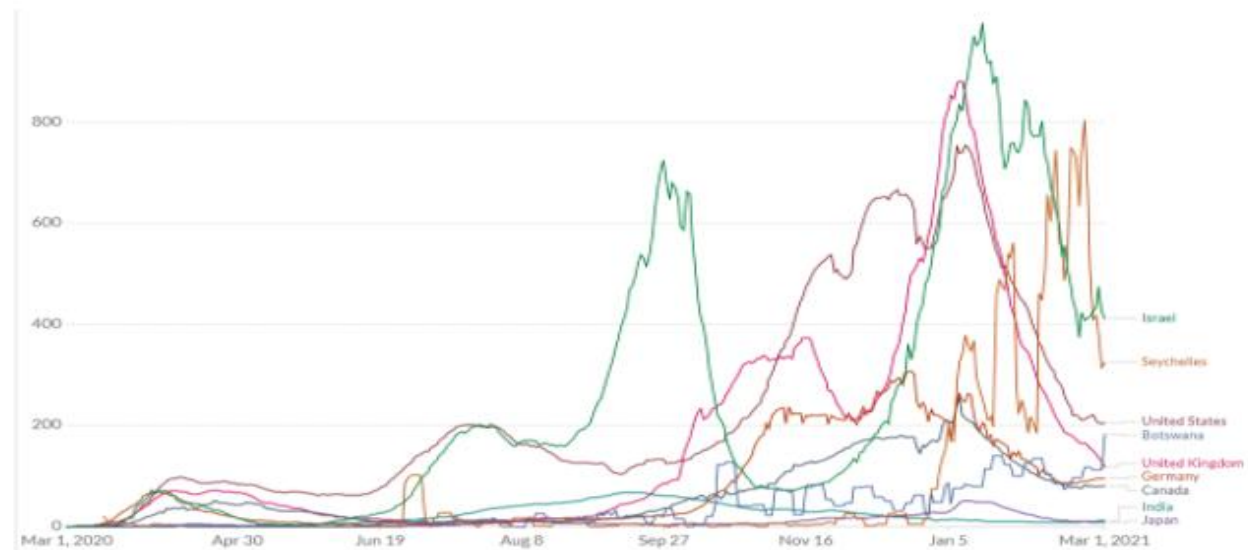
8. Statistics in Big Data and Data Science

- As we know that statistics are widely used in data analytics and data science technologies.
- But it is the foundation of Big data technologies too.
- Big data is nothing without the data, and data is useless without statistics.
- Thus Big Data technology totally depends on statistics.
- We use the statistics at the initial stage of Big data on the raw data sorting.
- Then we use it in other stages when we use Python programming in Big Data.
- We use Python in statistics to perform all primary functions of this technology

9. Statistics in the Health Industry

- Statistics is playing its part in the health industry. It helps the doctor to take and manage the data of their patients.
- Apart from this, WHO is also using statistics to generate its annual report about the health of the world's populations.
- Due to statistics, the medical scientist has invented lots of vaccines and anti tode to fight against major diseases.

- We have seen that in this COVID 19 Pandemic, the statistics are playing a crucial role in analyzing how much patience is all around the world, which region has the most number of cases and much more.
- Once the Covid-19 vaccine was developed, the government decided to vaccinate the people of their country. Below is the statistical data that define the number of people vaccinated in a particular country (till 1 March).
- Because of the statistical data, the government easily checks the number of people vaccinated in their country.



(This is the chart updated on 1 March 2021 of Corona patients)

Statistical organizations in India and their functions

1. Central Statistics Office (CSO):

- It is a governmental agency in India under the Ministry of Statistics and Programme Implementation responsible for co-ordination of statistical activities in India, and evolving and maintaining statistical standards. It has a well-equipped Graphical Unit. The CSO is located in Delhi.

- **Function:**

- The Central Statistics Office (CSO) in the Ministry of Statistics and Programme Implementation (MoS & PI) is responsible for the compilation of National Accounts Statistics (NAS).
- At the State level, State Directorates of Economy and Statistics (DEs) have the responsibility of compiling their State Domestic Product and other aggregates.
- It plays an advisory role in statistical matters.
- It provides national statistics to UN
- It has set up a unit to attend to statistical work relating to the five-year plans in collaboration with the planning commission and has expanded training facilities for statistics personnel.
- It is also responsible for the compilation and publication of national income statistics.
- The CSO through its Industrial Statistical wing conducts the Annual Survey of Industries and publishes the result.

2. Indian Statistical Institute (ISI)

It is a Civil Service and Research institute which is recognized as an Institute of National Importance by the 1959 act of the Indian parliament.

It grew out of the Statistical Laboratory set up by Prasanta Chandra Mahalanobis in Presidency College, Kolkata.

Established in 1931, this unique institution of India is one of the oldest institutions focused on statistics, and its early reputation led it to being adopted as a model for the first US institute of Statistics set up at the Research Triangle, North Carolina by Gertrude Mary Cox

Function:

- Primary activities of ISI are research and training of Statistics, development of theoretical Statistics and its applications in various natural and social sciences.

3. The National Sample Survey(NSS)

It is headed by a Director General is responsible for conduct of large scale sample surveys in diverse fields on All India basis.

Primarily data are collected through nation-wide household surveys on various socio-economic subjects, Annual Survey of Industries (ASI), etc.

Besides these surveys, NSS collects data on rural and urban prices and plays a significant role in the improvement of crop statistics through supervision of the area enumeration and crop estimation surveys of the State agencies.

It also maintains a frame of urban area units for use in sample surveys in urban areas.

Function:

- It conducts socio-economic surveys on various subjects like employment, consumer expenditure, health, unemployment, and medical services, etc.
- It decides the topics to be covered in a particular survey found.
- It conducts an annual survey of Industries every year.
- It gives every year reports on the status of estimation of agriculture production in India.

4. The International Institute for Population Sciences (IIPS)

It is formerly known as the Demographic Training and Research Centre (DTRC) till 1970, was established in July 1956 under the joint sponsorship of Sir Dorabji Tata Trust, the Government of India and the United Nations.

It serves as a regional centre for Training and Research in Population Studies for the ESCAP Economic and Social Commission for Asia and the Pacific region. The Institute was re-designated to its present title in 1985 to facilitate the expansion of its academic activities.

Case Study

- Case studies are in-depth studies of a phenomenon, like a person, group, or situation.
- The phenomenon is studied in detail, cases are analyzed and solutions or interpretations are presented.
- It can provide a deeper understanding of a complex topic or assist a person in gaining experience about a certain historical situation.
- Although case studies are used across a wide variety of disciplines, they are more frequently found in the social sciences.
- Case studies are a type of qualitative research.

Case Study.....contd

- It has been criticized as being unreliable, too general, and open to bias. To avoid some of these problems, studies should be carefully planned and implemented.
1. Determine the research question and carefully define it. The research question for case studies generally starts with a “How” or “Why.”
 2. Choose the cases and state how data is to be gathered and which techniques for analysis you’ll be using. Well designed studies consider all available options for cases and for ways to analyze those cases. Multiple sources and data analysis methods are recommended.
 3. Prepare to collect the data. Consider how you will deal with large sets of data in order to avoid becoming overwhelmed once the study is underway. You should formulate good questions and anticipate how you will interpret answers. Multiple collection methods will strengthen the study.
 4. Collect the data in the field (or, less frequently, in the lab). Collect and organize the data, keep good field notes and maintain an organized database.
 5. Analyze the data.
 6. Prepare your report.

TYPES OF DATA AND DATA CONDENSATION

Population:

Population is a collection of all the items under study about which we are trying to draw conclusions. A population is the complete set group of individuals, whether that group comprises a nation or a group of people with a common characteristic.. It refers to the set or group of all the units on which the findings of the research are to be applied. A population of interest may be the universe of nations or cities.

Example:

1. If one wants to study the academic performance of college all students of that college will constitute the population
2. If a company wants to market its new product in Maharashtra then the people in Maharashtra would be the population whose opinion will be sought

TYPES OF POPULATION:

- i. Finite and Infinite Population(On the basis of numbers)**
- ii. Homogeneous and Heterogeneous Population(On the basis of characteristics)**

i. FINITE AND INFINITE POPULATION

Finite Population:

The population in which number of units is finite and can be counted precisely, is called finite population.

Examples:

- Population of the students who appeared in S. S. C. Exam in Gujarat in March 2017.
- Students enrolled in Post Graduate courses in Gujarat.
- In service primary teachers of Rajasthan.

Infinite Population:

The population, in which the number of units is infinite and cannot be counted is called infinite population.

Examples:

- Decimal numbers lying between integers 1 and 2.
- Number of stars in the sky.
- Words formed by using all English alphabets.

ii. **HOMOGENEOUS POPULATION AND HETEROGENEOUS POPULATION**

Homogeneous Population:

If all the units of population are identical or similar in terms of certain characteristics, it is called homogeneous population.

Examples:

- **Blood in a living being** : Testing a single drop of blood reveals the condition of blood of whole body
- **Crop produced in a particular farm:** one can predict the quality of grain produced in a farm by checking a small quantity of grain ,

Heterogeneous Population:

If all the units of population differ completely or in some aspects with one another, the population is called heterogeneous population

Example:

- All students of the same school and class differ from one another in different mental and physical abilities

SAMPLE:

A Sample is a collection of some but not all the units of the population. It is a subset containing the characteristics of population. It means, the units, selected from the population as a sample, must represent all kind of characteristics of different types of units of population.

There are several procedures to decide on the units to be sampled namely Simple Random Sampling, Stratified Sampling, Cluster Sampling, Systematic Sampling etc.

SIMPLE RANDOM SAMPLING :

Simple random sampling is a process of selecting a sample of n units from a population of N units in a random manner such that each unit in the population has an equal chance of being included in the sample. A simple random sample is meant to be an unbiased representation of a group.

Example:

Suppose a firm has 1000 employees in which 100 of them have to be selected for onsite work. All their names will be put in a basket to pull 100 names out of those. In this case each employee has an equal probability of being included in the sample.

ADVANTAGES OF SIMPLE RANDOM SAMPLING(SRS):

- Ease of use represents the biggest advantage of simple random sampling.
- Unlike more complicated sampling methods, such as stratified random sampling and probability sampling, no need exists to divide the population into sub-populations or take any other additional steps before selecting members of the population at random.
- A simple random sample is meant to be an unbiased representation of a group. It is considered a fair way to select a sample from a larger population since every member of the population has an equal chance of getting selected.

DISADVANTAGES OF SIMPLE RANDOM SAMPLING(SRS):

- A sampling error can occur if the sample does not represent all the characteristics of the population

TYPES OF SIMPLE RANDOM SAMPLING :

- I. SIMPLE RANDOM SAMPLING WITHOUT REPLACEMENT
- II. SIMPLE RANDOM SAMPLING WITH REPLACEMENT

SIMPLE RANDOM SAMPLING WITHOUT REPLACEMENT(SRSWOR):'

In this method elements are drawn one by one without replacing the previous drawn element back in the population so that all the elements in the sample are distinct. Population size decreases after each draw is made. Suppose a sample of size n is to be selected from population of size N .

- i. For selection of the first sample unit all the N units are eligible for selection with equal chance i.e. All the units have an equal probability of $1/N$ of being included in the sample.
- ii. Now , the first unit selected is eliminated from the population leaving behind $(N-1)$ units
- iii. The second sample unit is now selected from remaining $(N-1)$ units in the population i.e. The probability of second sample unit being selected is $1/(N-1)$
- iv. The process continues until we have sample of n units

SIMPLE RANDOM SAMPLING WITH REPLACEMENT:

In this method elements are drawn one by one with the difference that each element drawn is put back in the population after noting down the observation, so that there can be repetitions of elements in the sample. The population size remains same at each draw.

A sample of size n is to be selected from a population of size N

- All the population units are eligible for inclusion every time a selection of sample unit is done.
- We do not drop the unit or units already selected.
- Thus all the population units have an equal probability $1/N$ everytime a sample is drawn
- Selected sample units may be repeated more than once

DATA:

- Data are a set of values of qualitative or quantitative variables about one or more persons or objects.
- Data are used in scientific research, businesses management (e.g., sales data, revenue, profits, stock price), finance, governance (e.g., crime rates, unemployment rates, literacy rates), and in virtually every other form of human organizational activity (e.g., censuses of the number of homeless people by non-profit organizations).
- Data are measured, collected, reported, and analyzed, and used to create data visualizations such as graphs, tables or images. Data as a general concept refers to the fact that some existing information or knowledge is represented or coded in some form suitable for better usage or processing

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- The background of the slide features a dark blue and black abstract design. On the left, there is a line graph with white circular markers connected by a white line. Below it, a bar chart with blue bars is visible. A specific data point on the bar chart is labeled with the number '289.33' in white. The right side of the slide is dominated by a large, light gray semi-circular shape that frames the text.
- **TYPES OF DATA:**
 - **Primary Data**
 - **Secondary Data**

Primary Data:

- Primary data are collected by the original(Direct) source. It is usually collected by the investigators himself or through his agents.
- They are original in character and are the raw materials of the enquiry
- There are different methods of collecting primary data such as interviews, surveys, experiments
- The researcher/investigator can choose particular method for data collection depending upon the research study
- **Example**
The population census conducted by the government of India after every ten years is primary data.

Interview Method:

- An interview is a method that involves two groups of people one is the interviewer who collects data or information and the other is the respondents who is being asked the questions by the interviewer
- The questions in the interview may be oral or verbal
- Interviews are carried out in 2 ways:
 - i. In-person Interview
 - ii. Telephonic Interviews
- An **In-person interview** requires an interviewer or a group of interviewers to ask questions from the respondents/interviewee in a face-to-face fashion. It can be direct or indirect, structured or unstructured, focused or unfocused, etc. Tools required for carrying out in-person interviews are notepad, recording device etc to keep note of the conversation
- Telephonic interviews are carried out over the phone through ordinary voice calls or video calls. The 2 parties involved may decide to use video calls like Skype to carry out interviews.

INTERVIEW METHOD:

ADVANTAGES

- ⑩ In-depth information can be collected.
- ⑩ Non-response and response bias can be detected.
- ⑩ The samples can be controlled.

DISADVANTAGES

- ⑩ It is more time-consuming.
- ⑩ It is expensive.
- ⑩ The interviewer may be biased.

SURVEYS AND QUESTIONNAIRES:

- Surveys and questionnaires are 2 similar tools used in collecting primary data. They are a group of questions typed or written down and sent to the sample of study to give responses.
- After giving the required responses, the survey is given back to the researcher to record. It is advisable to conduct a pilot study where the questionnaires are filled by experts and meant to assess the weakness of the questions or techniques used.
- There are 2 main types of surveys used for data collection,
 - i. online
 - ii. offline surveys.
- Online surveys are carried out using internet-enabled devices like mobile phones, PCs, Tablets, etc. They can be shared with respondents through email, websites, or social media.
- Offline surveys, on the other hand, do not require an internet connection for them to be carried out. The most common type of offline survey is a paper-based survey.
- However, there are also offline surveys like Formplus that can be filled with a mobile device without access to an internet connection. This kind of survey is called online-offline surveys because they can be filled offline but require an internet connection to be submitted.

SURVEYS AND QUESTIONNAIRES

Advantage:

- Respondents have adequate time to give responses.
- It is free from the bias of the interviewer.
- They are cheaper compared to interviews..

Disadvantage:

- A high rate of non-response bias
- It is inflexible and can't be changed once sent.
- It is a slow process.

OBSERVATION METHOD:

- The observation method is mostly used in studies related to behavioral science. The researcher uses observation as a scientific tool and method of data collection.
- Observation as a data collection tool is usually systematically planned and subjected to checks and controls.
- There are different approaches to the observation method—structured or unstructured, controlled or uncontrolled, and participant, non-participant, or disguised approach.
- The structured and unstructured approach is characterized by careful definition of subjects of observation, style of observer, conditions, and selection of data. An observation process that satisfies this is said to be structured and vice versa.
- A controlled and uncontrolled approach signifies whether the research took place in a natural setting or according to some pre-arranged plans. If an observation is done in a natural setting, it is uncontrolled but becomes controlled if done in a laboratory.

OBSERVATION METHOD

ADVANTAGES:

- ⑩ The data is usually objective.
- ⑩ Data is not affected by past or future events.

DISADVANTAGES:

- ⑩ The information is limited.
- ⑩ It is expensive.

FOCUS GROUPS:

- Focus Groups are gathering of 2 or more people with similar characteristics or who possess common traits. They seek open-ended thoughts and contributions from participants.
- A focus group is a primary source of data collection because the data is collected directly from the participant. It is commonly used for market research, where a group of market consumers engages in a discussion with a research moderator.
- It is slightly similar to interviews, but this involves discussions and interactions rather than questions and answers. Focus groups are less formal and the participants are the ones who do most of the talking, with moderators there to oversee the process.

FOCUS GROUP:

ADVANTAGES:

- ⑩ It incurs a low cost compared to interviews. This is because the interviewer does not have to discuss with each participant individually.
- ⑩ It takes lesser time too

DISADVANTAGES:

- ⑩ Response bias is a problem in this case because a participant might be subjective to what people will think about sharing a sincere opinion.
- ⑩ Group thinking does not clearly mirror individual opinions.

EXPERIMENT

- An experiment is a structured study where the researchers attempt to understand the causes, effects, and processes involved in a particular process.
- This data collection method is usually controlled by the researcher, who determines which subject is used, how they are grouped, and the treatment they receive.
- During the first stage of the experiment, the researcher selects the subject which will be considered. Therefore, some actions are carried out on these subjects, while the primary data consisting of the actions and reactions are recorded by the researcher.
- After which they will be analyzed and a conclusion will be drawn from the result of the analysis. Although experiments can be used to collect different types of primary data, it is mostly used for data collection in the laboratory.

EXPERIMENTS

ADVANTAGES:

- ⑩ It is usually objective since the data recorded are the results of a process.
- ⑩ Non-response bias is eliminated.

DISADVANTAGES:

- ⑩ Incorrect data may be recorded due to human error.
- ⑩ It is expensive.

ADVANTAGES OF PRIMARY DATA:

- **Specific:** Collecting your own data allows you the freedom to address issues specific to your business, or research aim. In this case, the data collected is exactly what the researcher wants and needs.
- **Accurate:** Primary data is much more accurate compared to secondary data. For example, when collecting statistical data from online sources, you are at risk of coming across false data.
- **Ownership:** The data collected through a primary source is usually owned by the researcher, who may choose to either share or not share with others.
- **Up to date information:** The data collected from primary sources are up-to-date, unlike that of the secondary sources. It collects data in real-time and does not take information from stale and outdated sources.
- **Control:** A researcher can easily control the research design and methods to be used. As a researcher, you can choose which subject to consider, and also control how the information is gathered.

Disadvantages of Primary Data:

1.Expensive

2.Time-consuming

3.Feasibility

Secondary Data Analysis:

- Secondary data analysis is the process of analyzing data collected from another researcher who primarily collected this data for another purpose.
- Researchers leverage secondary data to save time and resources that would have been spent on primary data collection.
- The secondary data analysis process can be carried out quantitatively or qualitatively depending on the kind of data the researcher is dealing with.
- The quantitative method of secondary data analysis is used on numerical data and is analyzed mathematically, while the qualitative method uses words to provide in-depth information about data.
- Different stages of Analyse Secondary Data:
 1. Statement of Purpose:
 - Before collecting secondary data for analysis, you need to know your statement of purpose. That is, a clear understanding of why you are collecting the data—the ultimate aim of the research work and how this data will help achieve it.
 - This will help direct your path towards collecting the right data, and choosing the best data source and method of analysis.

2. Research Design:

- This is a written-down plan on how the research activities will be carried out. It describes the kind of data to be collected, the sources of data collection, method of data collection, tools, and even method of analysis.
- A research design may also contain a timestamp of when each of these activities will be carried out. Therefore, serving as a guide for the secondary data analysis.
- After identifying the purpose of the research, the researcher should design a research process that will guide the data analysis process.

3. Developing the Research Questions:

- It is not enough to just know the research purpose, you need to develop research questions that will help in better identifying Secondary data. This is because they are usually a pool of data to choose from, and asking the right questions will assist in collecting authentic data.
- For example, a researcher trying to collect data about the best fish feeds to enable fast growth in fishes will have to ask questions like, What kind of fish is considered? Is the data meant to be quantitative or qualitative? What is the content of the fish feed? The growth rate in fishes after feeding on it, and so on.

4. Identifying Secondary Data:

- After developing the research questions, researchers use them as a guide to identifying relevant data from the data repository. For example, if the kind of data to be collected is qualitative, a researcher can filter out qualitative data.
- The suitable secondary data will be the one that correctly answers the questions highlighted above. When looking for the solutions to a linear programming problem, for instance, the solutions will be numbers that satisfy both the objective and the constraints.
- Any answer that doesn't satisfy both, is not a solution.

5. Evaluating Secondary Data

- This stage is what many classify as the real data analysis stage because it is the point where analysis is actually performed. However, the stages highlighted above are a part of the data analysis process, because they influence how the analysis is performed.
- Once a dataset that appears viable in addressing the initial requirements discussed above is located, the next step in the process is the evaluation of the dataset to ensure the appropriateness for the research topic. The data is evaluated to ensure that it really addresses the statement of the problem and answers the research questions.
- After which it will now be analyzed either using the quantitative method or the qualitative method depending on the type of data it is.

Advantages & Disadvantage of Secondary Data:

Advantage

- **Ease of Access :** Most of the sources of secondary data are easily accessible to researchers. Most of these sources can be accessed online through a mobile device.
- **Inexpensive:** Secondary data mostly require little to no cost for people to acquire them. Many books, journals, and magazines can be downloaded for free online.
- **Time-Saving:** The only investigation necessary for secondary data collection is the process of sourcing for necessary data sources.
- **Longitudinal and Comparative Studies:** Secondary data makes it easy to carry out longitudinal studies without having to wait for a couple of years to draw conclusions. For example, you may want to compare the country's population according to census 5 years ago, and now.
- **Generating new insights:** When re-evaluating data, especially through another person's lens or point of view, new things are uncovered. There might be a thing that wasn't discovered in the past by the primary data collector, that secondary data collection may reveal.

Disadvantage

- **Data Quality:** The data collected through secondary sources may not be as authentic as when collected directly from the source.
- **Irrelevant Data:** Researchers spend so much time surfing through a pool of irrelevant data before finally getting the one they need. This is because the data was not collected mainly for the researcher.
- **Exaggerated Data:** Some data sources are known to exaggerate the information that is being shared. This bias may be some to maintain a good public image or due to a paid advert.
- **Outdated Information:** Some of the data sources are outdated and there are no new available data to replace the old ones. For example, the national census is not usually updated yearly. Therefore, there have been changes in the country's population since the last census.

CROSS SECTIONAL DATA:

Cross-sectional data is the data that is collected by observing various subjects like(firms, countries, regions, people) at the same point of time, each observation belonging to a different individual. With cross-sectional data, we are not interested in the change of data over time, but in the current, valid opinion of the respondents about a question in a survey.

Example:

- i. Stock prices of 10 different IT companies in the year 2020
- ii. Sales of cars in different states in the month of June

TIME SERIES DATA:

- Time series data is the data that is collected by observing same elements or same variable at different points of time or different periods of time (hours, days, weeks, months, years etc.). This can be contrasted with cross-sectional data, which captures a point-in-time.
- The data is usually collected from the same variable but at specified times usually at equal intervals
- Time series are generally given in chronological order usually beginning with the earliest period

Example:

1. Profit of an organization over a period of 5 years.
2. Report of an heart attack patient taken after every 6 months.

Difference between time series data and cross sectional data

Time Series Data	Cross Sectional Data
1. A type of data consisting of observations of a single subject at multiple time intervals 2. Focuses on the same variable over a period of time 3. Eg. Profit of an organization over a period of 5 years time 4. It focuses on results gained over an extended period of time often within a small area 5. We analyze the change of variable over time	1. A type of data consisting of observations of many subjects at the same point in time 2. Focuses on several variables at the same point in time 3. Eg. Maximum temperatures of several cities on a single day 4. It focuses on information received from surveys and opinions at a particular time on various locations depending on the information needed 5. We analyze the relationship between different variables

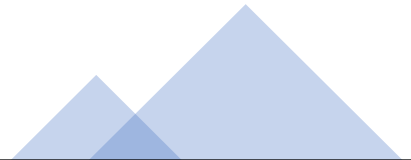
TABULATION



TABULATION :

The process of placing the classified data into tabular form is known as tabulation. A tabulation is logical and systematic arrangement of statistical data in rows and columns

Objectives of tabulation:

- To simplify complex data
 - To do away unnecessary details and save space
 - To make comparison easy
 - To facilitate statistical analysis
 - To identify trends and patterns of the data
- 

Requirements of Good Statistical Table(Components of Table)

Title : Each table must have a title displayed in bold type at the top of the table. It must be brief and self explanatory

Table Number : For easy reference of information in the table it must be numbered. The number can be put at the top above the title or at the bottom

Stubs and Captions : Stubs are row headings and captions are column headings. These heading should be well defined and brief.

Body of the table : It is the most important part of the table. It contains numerical information including meaningful row and column tools

Source note: It mentions source/origin of information used in the table

Foot Note: Normally placed at the bottom of the table, Footnotes are used to give explanation or to give further details about the symbols abbreviations used in the table

Table 1 ← Table no.

Title → **F.Y and S.Y. Admission Record of ABC College for 2008**

Stub ↓ Faculty	Class ← Caption								
	First Year			Second Year			Total		Grand Total
	Males	Females	Total	Males	Females	Total	Males	Females	
Arts	60	120	180	50	100	150	110	220	330
Science	100	66	166	90	58	148	190	124	314
Commerce	200	200	400	195	194	389	395	394	789
Total	360	386	746	335	352	687	695	738	1433

Source : Admission Files.

Figures updated till 4th June 2008

↑ – Source note

Body of the Table

↑ – Footnote

FORMS OF TABLE :

- 1. Simple and Complex Tables
- 2. General Purpose and Special Tables

i. Simple table :

When the data is classified with respect to one characteristic then the table is known as simple table or one way table

Following blank table may be used to show the number of students in a degree college for the year 2015-16 in different classes

Number of students in the college admitted in different classes for the year 2015-16

Class			
First Year	Second Year	Third Year	Total

ii. Complex Table:

When the data is classified with respect to two or more characteristics then the table is known as complex table

a. Two Way Table : It tabulates information related to two characteristics

Eg: Data is collected on number of students in the degree college for the academic year 2015-16 then we can tabulate the information according to classes and sex

Number of students in the college for the year 2015-16

SEX \ CLASS	First Year	Second Year	Third Year	Total

2. General and Special Purpose Table:

i. General Purpose Table:

Sometimes referred to as reference tables or information tables. These tables provide information for general use of reference. They usually contain detailed information and are not constructed for specific information.

Eg: Detailed tables prepared in census report

ii. Special Purpose Table:

These tables are known as summary tables which provide information for particular discussion. These tables are constructed or derived from general purpose tables. Useful for analytical and comparison studies involving study of relationship among variables

Eg: Calculation like ratios percentages etc

Tabulate the information :

In a town 60% are male, 80% of the male and 30% of female are non –vegetarians

Solution:

Distribution of Population in Town A with respect to sex and eating habit

Eating Habit	Sex		
	Male	Female	Total
Non-Vegetarians	48	12	60
Vegetarians	12	28	40
	60	40	100

FIGURES GIVEN IN PERCENTAGES

Example 2

From a specific area, 200 educated people were surveyed about their educational levels(UG,PG) and from which stream they belong to. 120 people were from Commerce stream, 60 were females and 150 were graduates. Of the commerce people 80 were graduates including 30 females. In all 40 females belong to commerce stream and among the females 48 were graduates. Construct a table with suitable title

Distribution of people sex wise and level of education in commerce and science stream

Sex ⇒ Stream \ Education level		Female	Male	Total
Commerce	G	30	50	80
	PG	10	30	40
Total		40	80	120
Science	G	18	52	70
	PG	2	8	10
Total		20	60	80
Total	G	48	102	150
	PG	12	38	50
Total		60	140	200

Foot note: G : Graduates

PG : Post Graduates

Example 4 :

An eatery in Chennai is frequented by both college students and employed males and females for dosa or idli. On a particular day 1800 customers visited the restaurant, 50% of them being males and 800 having dosa. 200 male students had dosa while two-third males had Idli of which 300 were employed males. 250 female students had dosa and 200 employed females had Idli. Present the above information in a suitable table.

Solution :

The information in the problem is based on three characteristics (i) Sex - Male or Female (ii) Status - Student or employed and (iii) Preference - Dosa or Idli. Thus we use a three-way table to present the data.

Table 6 : Customer Profile At An Eatery

Preference	Males			Females			Total		Grand Total
	Student	Employed	Total	Student	Employed	Total	Student	Employed	
Dosa	200	100	300	250	250	500	450	350	800
Idli	300	300	600	200	200	400	500	500	1000
Total	500	400	900	450	450	900	950	850	1800

DICHOTOMOUS VARIABLE:

Attribute that can be classified in any of 2 disjoint classes depending on whether the item possess the attribute or not is known as dichotomous or two fold classification

Notation and Terminology:

The capital Greek letters A,B,C are used to denote the names of various attributes and also to represent corresponding classes.

The Greek letters are used to denote the absence of attributes which are complements of classes A,B,C

N is the total number of items of population

A: Literate	α : Illiterate
B: Smoker	β : Non-smoker
C:Male	γ :Female

$AB\gamma$: A class of people who are literate females and are smokers

Class frequencies and order of classes:

Different attributes their subgroups and combinations are called different classes and the number of observations assigned to them are called class frequencies.

The frequencies of classes are denoted by putting corresponding combination of letters in a round bracket

Eg. In case of 2 attributes the number of classes are 9 i.e. $(A), (\alpha), (B), (\beta), (A\beta), (\alpha B), (AB)$ and N

Positive Frequencies: The frequencies of the type $(A), (AB), (ABC), \dots$ i.e. frequencies for the presence of attributes are known as positive frequencies.

Negative Frequencies: The frequencies of the type $(\alpha), (\beta), (\gamma), (\alpha\beta\gamma), \dots$ i.e. frequencies for the absence of attributes are known as negative frequencies

Contrary Frequencies: The frequencies of the type $(A\beta), (\alpha\beta C), \dots$ i.e. frequencies for presence of attributes in some and absence of attributes in other are known as contrary frequencies

Order of Classes :

- A class representing one attribute is called a class of first order.
Example: A, B, C, α , β , γ are classes of first order.
- A class representing two attribute is called a class of second order.
Example : AB, AC, A β etc. are classes of second order.
- Similarly, A β C, ABC, $\alpha\beta$ C are classes of third order.
- Thus we conclude order denotes the number of attributes in that class.

Class-Frequencies :

- The number of items belonging to a class is called the frequency of that class.
- The class frequency is denoted by putting the letter (or letters) denoting the class in a bracket. Thus, (A) stands for the number of items possessing the attribute A ; (α B) stands for the number of items, not possessing A and possessing B.
- The frequency of a positive class is called positive class frequency e.g. (AB) and frequency of a negative class is called negative class frequency e.g. ($\alpha\beta\gamma$).

ORDER OF CLASSES:

The class having n attributes refers to the class that has nth order i.e. order of the class is the number of attributes present in it .

Eg. (AB) refers to class of 2nd order because of presence of 2 attributes i.e. A and B

ORDER OF CLASS	FREQUENCIES
0	N
1	(A),(B),(α),(β)
2	(AB),(Aβ),(αB),(αβ)

FOR 3 attributes A,B,C the frequencies of different orders

Order	Frequencies
0	N
1	(A),(B),(C),(α),(β),(γ)
2	(AB),(A α),(A β),($\beta\alpha$)
3	(ABC)(A β C)(AB γ)(α BC) ($\alpha\beta$ C) (α B γ) ($\alpha\beta\gamma$)

Ultimate Class frequencies:

- The class-frequencies of highest order are called ultimate class-frequencies. Thus, in the case of two attributes class-frequencies of order two are ultimate class-frequencies.
- If A and B are attributes then (AB), (A β), (α B), ($\alpha\beta$) are ultimate class-frequencies.
- If we are considering n attributes, the Ultimate class frequencies will have n symbols.
- Thus the total number of ultimate class frequencies in case of two attributes are $2^2 = 4$ and for three attributes are $2^3 = 8$
- The total number of ultimate class frequencies in case of “n” attributes are 2^n
- The total number of positive class frequencies are 2^n
- The total number of class frequencies of all order are 3^n

Contingency Table: Frequency data for attributes can be presented in a tabular form called contingency table

CONTINGENCY TABLE FOR 2 ATTRIBUTES

	A	α	TOTAL
B	(AB)	(A α)	(B)
β	(A β)	($\beta\alpha$)	(β)
TOTAL	(A)	(α)	N

CONTINGENCY TABLE FOR 3 ATTRIBUTES

		C	γ	TOTAL
A	B	(ABC)	(AB γ)	(AB)
	β	(A β C)	(A $\beta\gamma$)	(A β)
	TOTAL	(AC)	(A γ)	(A)
α	B	(α BC)	(α B γ)	(α B)
	β	($\alpha\beta$ C)	($\alpha\beta\gamma$)	($\alpha\beta$)
	TOTAL	(α C)	($\alpha\gamma$)	(α)
TOTAL	B	(BC)	(B γ)	(B)
	β	(β C)	($\beta\gamma$)	(β)
	TOTAL	(C)	(γ)	N

The following are class frequencies for a certain data on 3 attributes A , B and C . Put these frequencies on the contingency table and complete the table. Also comment whether given data is consistent. $N = 1000$, $(A) = 88$, $(B) = 109$, $(C) = 28$, $(AB) = 34$, $(BC) = 13$, $(AC) = 14$, $(ABC) = 6$.

Attribute		C	γ	Total
A	B	6	$28 = (34 - 6)$ ②	34
	β	$8 = (14 - 6)$ ①	$46 = (74 - 28)$ ⑤	$54 = (88 - 34)$ ③
	Total	14	$74 = (88 - 14)$ ④	88
α	B	$7 = (13 - 6)$ ⑪	$68 = (96 - 28)$ ⑬	$75 = (7 + 68)$ ⑮
	β	$7 = (15 - 8)$ ⑬	$830 = (898 - 68)$ ⑮	$837 = (7 + 830)$ ⑮
	Total	$14 = (28 - 14)$ ⑫	$898 = (912 - 14)$ ⑮	$912 = (1000 - 88)$ ⑭
Total	B	13	$96 = (109 - 13)$ ⑥	109
	β	$15 = (28 - 13)$ ⑦	$876 = (972 - 96)$ ⑩	$891 = (1000 - 109)$ ⑧
	Total	28	$972 = (1000 - 28)$ ⑨	1000

Example 4:

To study the popularity of ice cream flavor, a survey was conducted of 450 people from a particular area. Number of people who like chocolate, vanilla and butterscotch flavor are 265, 242 and 250 respectively. 135 like both chocolate and vanilla, 132 people like both chocolate and butterscotch and 144 like both vanilla and butterscotch. The number of people who like all three flavors is 88. Find the number of people who like at least two flavors.

Solution:

A : people who like chocolate B : people who like vanilla C : people who like butterscotch

$N = 450$, $(A) = 265$, $(B) = 242$, $(C) = 250$, $(AB) = 135$, $(AC) = 132$, $(BC) = 144$, $(ABC) = 88$

The information can be represented in tabular form as follows.

		C	γ	Total
A	B	$(ABC) = 88$	$(AB\gamma) = \underline{47}$	$(AB) = 135$
	β	$(A\beta C) = \underline{44}$	$(A\beta\gamma) = \underline{86}$	$(A\beta) = \underline{130}$
	Total	$(AC) = 132$	$(A\gamma) = \underline{133}$	$(A) = 265$
α	B	$(\alpha BC) = \underline{56}$	$(\alpha B\gamma) = \underline{51}$	$(\alpha B) = \underline{107}$
	β	$(\alpha\beta C) = \underline{62}$	$(\alpha\beta\gamma) = \underline{16}$	$(\alpha\beta) = 78$
	Total	$(\alpha C) = \underline{118}$	$(\alpha\gamma) = \underline{67}$	$(\alpha) = \underline{185}$
Total	B	$(BC) = 144$	$(B\gamma) = \underline{98}$	$(B) = 242$
	β	$(\beta C) = \underline{106}$	$(\beta\gamma) = \underline{102}$	$(\beta) = \underline{208}$
	Total	$(C) = 250$	$(\gamma) = \underline{200}$	$N = 450$

Number of people who like at least two flavors $= (AB\gamma) + (A\beta C) + (\alpha BC) + (ABC)$
 $= 47 + 44 + 56 + 88 = 235$

Consistency of Data:

- If all the class frequencies are positive then the data is said to be consistent. Or if all the ultimate class frequencies are non-negative then the data is consistent.
- Conditions of Consistency

1. For ONE attribute:

(i) $(A) > 0,$

(ii) $(\alpha) > 0.$

(iii) $(A) \leq N,$

(iv) $(\alpha) \leq N$

But $(A) + (\alpha) = N$

2. For TWO attribute:

$$(i) N = (A) + (\alpha) = (B) + (\beta)$$

$$(ii) (A) = (AB) + (A\beta), \text{ and } (B) = (AB) + (\alpha B)$$

$$(\alpha) = (\alpha B) + (\alpha\beta), \text{ and } (\beta) = (A\beta) + (\alpha\beta)$$

$$(AB) \geq 0$$

$$(A\beta) \geq 0$$

$$\text{Let } (A\beta) = AB \cdot N$$

$$\beta = 1 - B$$

$$\therefore AB \cdot N \geq 0$$

$$A(1-B) \cdot N \geq 0$$

$$AN - AB \cdot N \geq 0$$

$$(A) - (AB) \geq 0$$

$$(A) \geq (AB)$$

$$(i) (\alpha B) \geq 0$$

$$\alpha B \cdot N \geq 0$$

$$\text{Let } (\alpha B) = \alpha B \cdot N$$

$$\alpha = 1 - A$$

$$(1-A)B \cdot N \geq 0$$

$$B \cdot N - AB \cdot N \geq 0$$

$$(B) - (AB) \geq 0$$

$$(B) \geq (AB)$$

$$(AB) \geq 0$$

$$AB \cdot N \geq 0$$

$$(1-A)(1-B) \cdot N \geq 0$$

$$N - A \cdot N - B \cdot N + AB \cdot N \geq 0$$

$$(AB) \geq (A) + (B) - N$$

$$\text{Let } (AB) = AB \cdot N$$

$$\alpha = 1-A \quad \beta = 1-B$$

Consistency conditions for TWO attributes:

- (i) $(AB) \geq 0$.
- (ii) $(\alpha\beta) \geq 0$ $(A\beta) \geq 0$.
- (iii) $(\alpha B) \geq 0$
- (iv) $(\alpha\beta) \geq 0$
- (v) $(A) \geq (AB)$.
- (vi) $(B) \geq (AB)$
- (vii) $(AB) \geq (A) + (B) - N$

Consistency conditions for THREE attributes:

(i) $(ABC) \geq 0$.

(ii) $(AB) \geq (ABC)$

(iii) $(AC) \geq (ABC)$

(iv) $(BC) \geq (ABC)$

(v) $(ABC) \geq (AB) + (AC) - (B)$

(vi) $(ABC) \geq (AB) + (AC) - (A)$

(vii) $(ABC) \geq (AC) + (BC) - (C)$

(viii) $(ABC) \leq (AB) + (BC) + (AC) - (A) - (B) - (C) + N$

Example :

From the following data check whether the data are consistent or not. $(A) = 120$, $(B) = 165$, $(AB) = 160$, $N = 400$.

Solution :

$$(\alpha B) = (B) - (AB) = 165 - 160 = 5,$$

$$(\alpha) = N - (A) = 400 - 120 = 280$$

$$(\alpha\beta) = (\alpha) - (\alpha B) = 280 - 5 = 275$$

$$(A\beta) = (A) - (AB) = 120 - 160 = -40$$

Since one of the ultimate class frequency is negative the given data is not consistent.

Association of Attributes :

- To study relationship if the characteristics cannot be measured i.e. to study relationship between two attributes we use the technique called association of attributes.
 - If the attributes are not independent and they are related with each other in some way then they are said to be associated to one another.
1. Positive Association: Two attributes are said to be positively associated when they are present or absent together . In mathematical terms; i.e. $(AB) > (A)(B) / (N)$ then A & B are Positively Associated.
 2. Negative Association: Two attributes are said to be negatively associated if combination of presence of one attribute with absence of other attribute appears more frequently. In mathematical terms i.e. $(AB) < (A)(B) / (N)$ then A & B are Negatively Associated.
 3. A always occurs with B and never with β i.e. $(AB)=(A)$ or $(A \beta)=0$ or B always occurs with A and never with α i.e. $(AB)=(B)$ or $(\alpha B)=0$ then A and B are completely associated
 4. On the other hand if A never occurs with B i.e. $(AB)=0$ or α never occurs with β i.e. $(\alpha \beta)=0$

A group of 200 students was studied with respect to success in an examination. Out of 120 boys 80 passed the examination. Of the girls, only 10 failed. Present this information in a 2×2 contingency table and comment on the type of association.

Solution :

Let A : boys; α : girls

B : Pass; β : fail

Thus the given frequencies are

$N = 200$; $(A) = 120$; $(AB) = 80$; $(\alpha\beta) = 10$

The corresponding 2×2 contingency table will be

	B	β	Total
A	$(AB) = 80$	$(A\beta) = 40$	$(A) = 120$
α	$(\alpha B) = 70$	$(\alpha\beta) = 10$	$(\alpha) = 80$
Total	$(B) = 150$	$(\beta) = 50$	$N = 200$

$$\text{Now, } \frac{(A)(B)}{N} = \frac{120 \times 150}{200} = 90$$

$$\therefore (AB) < \frac{(A)(B)}{N}$$

$\therefore A$ and B are negatively associated.

Measures of Association

1. Yule's Coefficient of association:

- This is the most commonly used method of studying association and it is denoted by Q and is defined by ;

$$Q = \frac{(AB)(\alpha\beta) - (A\beta)(\alpha B)}{(AB)(\alpha\beta) + (A\beta)(\alpha B)}$$

- If A and B are independent then $(AB)(\alpha\beta) - (A\beta)(\alpha B) = 0$, If A and B are independent then $(AB)(\alpha\beta) - (A\beta)(\alpha B) = 0$, hence $Q = 0$.
- If A and B are completely associated, then $(AB) = (A)$ and $(AB) = (B)$
i.e. $(A\beta) = 0$ and $(\alpha B) = 0$. Thus $Q = 1$
- If A and B are completely dissociated, then $(AB) = 0$ & $(\alpha\beta) = 0$ Thus $Q = -1$
i.e. Q lies between -1 to 1.

2. Coefficient of Colligation :

- Another measure of association coefficient suggested by Yule is the coefficient of colligation denoted by Y and is given by ;

$$Y = \frac{1 - \sqrt{\frac{(A\beta)(\alpha B)}{(AB)(\alpha\beta)}}}{1 + \sqrt{\frac{(A\beta)(\alpha B)}{(AB)(\alpha\beta)}}}$$

- If A & B are independent then $(AB)(\alpha\beta) = (A\beta)(\alpha B)$

Thus $Y = 0$

- If A and B are completely associated, then $(AB) = (A)$ and

$(\alpha B) = (B)$ i.e. $(A\beta) = 0$ and $(\alpha\beta) = 0$. Thus $Y = 1$

- If A and B are completely dissociated, then $(AB) = 0$ & $(\alpha\beta) = 0$ Thus $Y = -1$

i.e. Y lies between -1 to 1.

Questions

1. When $(AB)=(A)$ and $(AB)=(B)$ then what kind of association it is?
2. Give Yules Coefficient of Association in each of the following case?
 - i. If there is perfect association between 2 attributes
 - ii. If coefficient of colligation is 0.8
 - iii. If every individual that possess an attribute A also possess the attribute B?
 - iv. When 2 attributes are completely dissociated

1800 candidates appeared for an examination. 1350 were successful, 400 had attended a coaching class and out of these 50 came out successful. Compute the Yule's coefficient of association and comment on your findings.

Solution :

Let A : successful candidates

α : failed candidates

B : candidates who attended coaching classes

β : candidates who did not attend coaching classes

From the information in problem

$N = 1800$; $(A) = 1350$; $(B) = 400$; $(AB) = 50$

	B	β	Total
A	$(AB) = 50$	$(A\beta) = 1300$	$(A) = 1350$
α	$(\alpha B) = 350$	$(\alpha\beta) = 100$	$(\alpha) = 450$
Total	$(B) = 400$	$(\beta) = 1400$	$N = 1800$

$$\begin{aligned}
 Q &= \frac{(AB)(\alpha\beta) - (A\beta)(\alpha B)}{(AB)(\alpha\beta) + (A\beta)(\alpha B)} \\
 &= \frac{50 \times 100 - 350 \times 1300}{50 \times 100 + 350 \times 1300} = \frac{500 - 455000}{500 + 455000} = \frac{-454500}{455500} \\
 &= -0.9978
 \end{aligned}$$

The high negative value of Q suggests that coaching classes are not useful success.